Step 1: corrected U(1) background test

UBT

2026-10-31

AI provenance — Tier B. AI-assisted; machine-verified against the named verifier; individual human attestation is not claimed.

1 Scope and corrected status

GAP-U1-NEUTRALITY: FORMER PROOF REFUTED; PHYSICAL BRIDGE OPEN. The former argument in `canonical/qed_phi_const/step1_u1_protection.tex` replaced the declared left phase action by conjugation. We correct that argument and continue only on the original single-biquaternion carrier. The microscopic action, physical charge representation and constrained positive fluctuation spectrum remain unselected. No new field, metric or universe identification is adopted.

2 The action, not commutation, determines charge

For a psi-independent electromagnetic gauge parameter, every Fourier coefficient has the same declared left charge. Being real or having zero Fourier index does not make a nonzero field invariant. Central conjugation would leave the ENTIRE algebra invariant and could not charge just the other components. The old theorem therefore establishes neither neutrality nor a massless photon.

$$\Theta' = e^{i\alpha}\Theta, \qquad \frac{d}{d\alpha}(e^{i\alpha}v)\Big|_{\alpha=0} = iv \neq 0 \quad (v \neq 0),$$
$$e^{i\alpha}ve^{-i\alpha} = v. \qquad \theta'_n = e^{i\alpha}\theta_n.$$

3 Conditional gauge-boson mass test

Assume a positive canonically normalized complex-scalar kinetic term, Hermitian charge Q and a constant background v, with natural units and signature (+—). Comparing its quadratic gauge term with the Proca normalization gives the following result. Here v is the complex field value itself, not the real radial convention v divided by the square root of two. Thus a nonzero background in the declared charge-one representation fails the neutral-background test. This is a diagnostic of the former scalar comparator, not a derivation of the unresolved UBT physical Hessian. Gauge invariance of an action alone does not imply a massless gauge boson.

$$\mathcal{L}_{\text{kin}} = (D_\mu\phi)^\dagger D^\mu\phi, \qquad D_\mu = \partial_\mu + ieQA_\mu,$$
$$\mathcal{L}_{A^2} = e^2v^\dagger Q^2 vA_\mu A^\mu = \tfrac{1}{2}m_\gamma^2 A_\mu A^\mu,$$
$$m_\gamma^2 = 2e^2\|Qv\|^2, \qquad m_\gamma^2 = 0 \Longleftrightarrow Qv = 0 \quad (e \neq 0).$$

4 Verification and continuation

The analytical correction and independent checks are recorded in the paired research note `u1_neutrality_composite_channel` and `reports/u1_neutrality_2026_10_31.json`. The verifier `tools/verify_u1_neutrality.py` checks the phase counterexample and positive-scalar mass normalization. LEAN-PENDING: neither lean nor lake is installed. This does not derive a UBT physical Hessian or a multiverse channel. The former QED conclusion is OPEN, not PROVED. English is the translation source. Independent semantic translation approval and author attestation are not claimed.